Section 1 : ISRO Intro and History

Learn about : INCOSPAR, DAE to DOS, Dr Sarabhai, SITE, Coconut Root Wilt, Rice Estimates, Global Village, Church, presentation before PM, Start within few years of Independence so social upliftment as main objective, Application oriented organisation, Also tell who control ISRO .

Cover these topics using Document which I shared , Wikpedia , Google Gemini, Youtube, ChatGPT . Clear every Point . Compare with NASA Formation .

- Refrence : Document which I shared , Wikpedia , Google Gemini, Youtube, ChatGPT .
- Create PPT, Animated Video . Prepare also word file of it . Learn every point .
- Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used

Section 2 : ISRO History & Centres

What is ISRO ? How many Centres ? Where located ? What activities carried out ? Where is Head Quarter ? About Chairman and Directors of each centre . About ISTRAC and Research Laboratories also ?

Also Tell about Commercial Centres like INSPACE and their purpose ? Which Ministry control ISRO ? Scope of Career in ISRO Centres for every stream? How you can get Job in ISRO Centres – Eligibility , Selection , Payscale , Post

Cover these topics using Document which I shared , ISRO Website , ChatGPT . Clear every Point .

- **Refrence : Document which I shared , ISRO Website , Website of each ISRO Centres , ChatGPT .**
- **Create PPT, Animated Video . Prepare also word file of it . Learn every point .**
- **Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used**

Section 3 : Initial Satellites :

Initial Days of ISRO, Experimental Satellites, Spin Stabilized, single channel, km resolution, launch from USSR

Learn about : Initial Days of ISRO, About Satellites : What is Satellite ? what are the types of satellite ? Experimental Satellites, Spin Stabilized, single channel, km resolution, launch from USSR . Total Satellites of India Yet : 108 . What are space debris ? Total space debris tracked : 35017. About all things of these initial satellites Aryabhatt, Bhaskar and APPLE & SROSS-Weight,Payloads , When Launch, From where ? Which Launching Vehicle and of which country , Purpose of them , Lifetime, NORAD ID , Orbital Period, Perigee, Apogee, Orbit , How far from earth ? What experiments or work carried out ? when got inactive or dead ? what is current location now ? What is Current Status ? Orbital Parameters .

Cover these topics using Document which I shared , ISRO Website , ChatGPT , Wikpedia , N2YO, Clear every Point .

- **Refrence : Document which I shared , ISRO Website , ChatGPT , Wikpedia , N2YO, Clear every Point .**
- **Create PPT with Images, Animated Video . Prepare also word file of it . Learn every point .**
- **Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used**

Section 4 : Orbits of Satellites

Learn about : Satellite , Satellite Orbits , Keplers Law, Orbital Parameters Arthur Clarke, Types of orbits, Advantage of Height, Polar orbits for RS self scanning, Sun Sync for same illumination, Geostationary for Comm to make it always visible, 3 Sat to cover world, Arthur Clarke

Types of Orbit- LEO,MEO, GEO, SSO, Polar Orbit, Elliptical, Inclined- Learn all of them, their use and why they use ? Height, Orbital Period, Visibility Time, Advantage of Height , Which Satellites in each type of orbit : Name some example , Sunsynchronous

Cover these topics using Document which I shared , ChatGPT , Wikpedia , N2YO, Books (Online) Clear every Point .

- **Refrence : Document which I shared , ISRO Website , ChatGPT , Wikpedia , N2YO, Clear every Point .**
- **Create PPT with Images, Animated Video . Prepare also word file of it . Learn every point .**
- **Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used**

Section 5 : Remote Sensing

Learn about : What is Remote Sensing ? How and Why? Eye vs Camera, Sunlight, Spectral Signature, Passive & Active, Height advantage of photo, km to cm resolution, Advantage of Remote Sensing . About Earth Observation satellites – Why , How and Applications. Best commercial satellite in world related to Remote Sensing Data specially Indian within few years – IRS 1C, data sold to many organizations & commercial satellite in world related to Remote Sensing Data of NASA . Compare NASA and India in terms of Remote Sensing , Best commercial satellite, Features, Advantages, Cost, Applications

Learn about : RISAT, Resourcesat, Oceansat, Cartosat- their upgraded version like Cartosat, Cartosat 2 , Cartosat 3, Resource Sat 2 ,EOS -1, 2 and so on. About these satellites learn :

Weight,Payloads , When Launch, From where ? Which Launching Vehicle and of which country , Purpose of them , Lifetime, NORAD ID , Orbital Period, Perigee, Apogee, Orbit , How far from earth ? What experiments or work carried out ? when got inactive or dead ? what is current location now ? What is Current Status ? Orbital Parameters .

- **Refrence : Document which I shared , ISRO Website , ChatGPT , Wikpedia , N2YO, Clear every Point .**
- **Create PPT with Images, Animated Video . Prepare also word file of it . Learn every point .**
- **Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used**

Section 6 : Remote Sensing Applications

Prepare about : Application of Remote Sensing , Name of Areas where it used . Explanation of Remote Sensing in various Areas like Agriculture, Meterology, Ocean, Hydrology , Fishing, Urban Planning etc.

Name of Satellites of India and NASA with their details and payloads & their role & Application Explanation related to

- Fishing – all budget recovered
- Agri Estimates
- Urban planning
- Geology- Water level
- Land Usage
- Soil Types
- Glaciers
- Ocean Science
- Meterlogical- Weather and Clouds – Cloud Burst

- Forest & Environment
- Geology and Geomorphology
- E-Governance
- Land Resources Rural Development
- Urban Development
- Water Resources
  - Bore well using Rajiv Gandhi National Drinking Water scheme
  - Efficiency more than 90%
- Weather and Climate
- National Meet Outcomes
- Capacity Building
- Solar Calculator and other Apps on Vedas Portal

Learn about Bhuvan, Vedas & MOSDAIC Portal in details--What Data they give and How in which format. How they are used ? Demo of Data shown ?

Also tell facts where they are beneficial give examples of Indian related to past activities or events .

- **Reference - Learn from : Documents Uploaded, ISRO Websites , ISRO Websites- About Satellites of Remote Sensing in India , Vedas, Bhuvan and MOSDAIC Portal , News related to past events applications . Internet, Wikipedia , some about Chat GPT – In Chat GPT use IRO Official word in prompt.**
- **Prepare PPT in which slides of each topic with picture , Animated Video . Prepare also word file of it . Learn every point .**
- **Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used**

Section 7 : Communication Satellite & Applications

Learn about :

Communication- What, How and Why Sat Comm

Components used in Communication Satellites

Line of sight, Bands used

5 slots of India

Name of Satellites of India and NASA with their details and payloads & their role & Application Explanation related to

- GSAT & EDUSAT Series, IRNSS E Learning in details
- 2 to 1000+ channels progress Satellite Phones
- DTH – TV
- Weather
- 24x7 Services like ATM
- VSAT
- Navigation

Explanation related to Application in

- Saving lives – Lakhs to 100s death due to cyclones
- High Throughput,
- DAT: Low cost, floats on water
- few buttons so very simple to operate, fast response, location and type of emergency sent
- Also tell facts where they are beneficial give examples of Indian related to past activities or events or news.

➢ Reference - Learn from : Documents Uploaded, ISRO Websites , ISRO Websites- About Communication Satellites of India- Name of Satellites of India- GSAT, EDUSAT, DAT in details with payload – GSAT & EDUSAT

Series, IRNSS E Learning, News related to past events applications . Internet, Wikipedia , some about Chat GPT – In Chat GPT use IRO Official word in prompt.

- **Prepare PPT in which slides of each topic with picture , Animated Video . Prepare also word file of it . Learn every point .**
- **Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used**

Section 8 : Navigation Applications

Learn about : What is Navigation ? Which satellites are used at what height ,which orbit ? how many satellites used in Navigation ? US Satellites for Navigation , Indian satellites for Navigation ? Name and their details – Launching, Orbit , Payloads .

How Navigation principle by satellite work ?

what signals are sent by user and what signals sent by satellite for Navigation ?

Applications and Working Concept of :

- Nav ki Gati
- Space – Ground – User segments
- 4 equations-4 Unknowns
- How it IRNSS (Indian Navigation) better than GPS others ?
- Application :
- Vehicle Tracking, Railway crossing, ship movement, Aircraft tracking, NavIC
- available in MI 8
- Which Mobiles support Navigation by Indian satellites
- Which phones support Navic ?

- Highly accurate clock
- Features of Navic
- About Navic App
- How it helps Fisherman

- **Reference - Learn from : Documents Uploaded, ISRO Websites , ISRO Websites- About Navigation Satellites of India - Name of Satellites of India- IRNSS and Navic in details with payload – IRNSS Series,Vedas Portal ,Navic App . Internet, Wikipedia , some about Chat GPT – In Chat GPT use ISRO Official word in prompt.**
- **Prepare PPT in which slides of each topic with picture , Animated Video . Prepare also word file of it . Learn every point .**
- **Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used**

Section 9: Chandrayaan 1 & 2

Learn about :

- What is Moon ? How it create ? How far Moon ?
- Why Moon study ?
- What we are searching on Moon ?
- How to Reach Moon ?
- How NASA, Russia, China reach Moon where, when, which rockets used ? in How many days
- India's First attempt – Chandrayaan 1
- Why Chandrayaan 1 ?
- What we are studying on Moon through Chndrayaan 1 ?
- International Mission with 6 foreign and 5 ISRO instruments,
-

- Tell about No. of years to prepare, Total Budget, When, Details of Launching Vehicle, Trajectory, Days, Payloads of Chandrayaan and their role , what are their experiments and images carried out ?
- Name of Scientist, Mission Director, of Chandrayaan 1, Payload Designer, Where they developed ?
- Gallery of Chandrayaan 1 Images from ISRO
- From where Data of Chandrayaan 1 found ? ISDDC Portal, What Data it give in which format ?
- What Experiments Carried out – Images
- Curent Status

- India's Second attempt – Chandrayaan 2
- Tell about When, Details of Launching Vehicle, Trajectory, Days, Payloads of Chandrayaan 2 and their role , what are their experiments and images carried out ?
- Trajectory
- Name of Scientist, Mission Director, of Chandrayaan 1, Payload Designer, Where they developed ?
- Gallery of Chandrayaan 2 Images
- From where Data of Chandrayaan 2 found ? ISDDC Portal, What Data it give in which format ?
- What Experiments Carried out – Images
- Water on Moon, 3D maps
- Failure and Reason
- Curent Satus
- Compare Chandrayaan 1 & 2.
- Compare Mission of Moon of NASA, China, India and Russia?

- **Reference - Learn from : Documents Uploaded, ISRO Websites , ISRO Websites- About Chandrayaan 1 & 2 in details with payload , Launching Vehicle Details – ISDDC Portal and ISRO Website and ISRO Gallery . Internet, Wikipedia , some about Chat GPT – In Chat GPT use IRO Official word in prompt.**
- **Prepare PPT in which slides of each topic with picture , Animated Video . Prepare also word file of it . Learn every point .**
- **Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used**

Section 10 – Chandrayaan 3

Learn about :

- Studied Carried out by Chandrayaan 1 & 2
- Failure of Chandrayaan 2
- Why Chandrayaan 3 ? Whats' aim ? How we tried to achieve ?
- What we are studying on Moon through Chndrayaan 3 ?
- How NASA, Russia, China Rover reach on Moon where, when, which rockets used ? in How many days
- International Mission with 6 foreign and 5 ISRO instruments,
- Tell about No. of years to prepare, Total Budget, When, Details of Launching Vehicle, Trajectory, Days, Payloads of Chandrayaan and their role , what are their experiments and images carried out ?
- Name of Scientist, Mission Director, of Chandrayaan 1, Payload Designer, Where they developed ?
- Landing Location , Pole of Moon, Cordinates, Name of Point , Time Lines
- How many Days Carried out ? Why we select 23 August ?
- Gallery of Chandrayaan 3 Images from ISRO

- From where Data of Chandrayaan 3 found ? ISDDC Portal, What Data it give in which format ?
- What Experiments Carried out – Images
- Water on Moon, 3D maps, seismic activity, Temperature, Minerals and other things and experiments done with Data and Graph
- Curent Status
- Compare Chandrayaan 1 & 2 & 3.
- Compare Mission of Moon of NASA, China, India and Russia?

➢ **Reference - Learn from : Documents Uploaded, ISRO Websites , ISRO Websites- About Chandrayaan 3 in details with payload , Launching Vehicle Details – ISDDC Portal and ISRO Website and ISRO Gallery . Internet, Wikipedia , some about Chat GPT – In Chat GPT use IRO Official word in prompt.**

➢ **Prepare PPT in which slides of each topic with picture , Animated Video . Prepare also word file of it . Learn every point .**

➢ **Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used**

Section 11 : Mangalyaan (Mars Orbiter Mission)

Learn about :

- What is Mars ? How it create ? How far Mars ?
- Why Mars study ?
- What we are searching on Mars ?
- How to Reach Mars ?How many Days ? Why we select that day to Launch MOM?
- How NASA, Russia, China reach Mars where, when, which rockets used ? in How many days

- India's First attempt – MOM
- Why MOM ?
- What we are studying on Mars through MOM ?
- Tell about No. of years to prepare, Total Budget, When, Details of Launching Vehicle, Trajectory, Days to reach , Payloads of MOM and their role , what are their experiments and images carried out ?
- First attempt success, Map, 12ppb Methane Sensor, 50km error in 50 Cr Km, 300 days travel

- Name of Scientist, Mission Director, of MOM, Payload Designer, Where they developed ?
- Gallery of MOM Images from ISRO
- From where Data of MOM found ? ISDDC Portal, What Data it give in which format ?
- What Experiments Carried out – Images
- Curent Status

➢ **Reference - Learn from : Documents Uploaded, ISRO Websites , ISRO Websites- About Mangalyaan in details with payload , Launching Vehicle Details – ISDDC Portal and ISRO Website and ISRO Gallery . Internet, Wikipedia , some about Chat GPT – In Chat GPT use IRO Official word in prompt.**

➢ **Prepare PPT in which slides of each topic with picture , Animated Video . Prepare also word file of it . Learn every point .**

➢ **Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used**

Section 12 : GaganYaan

Learn about :

- What is Man oin Space Mission ? What is Gaganyaan
- Why Gaganyaan ?
- What we will do with Gaganyaan ?
- Number of Persons, who they are , which background, what are eligibility criteria of selection, where they are getting training ? why they selected?
- Mission Details- when start, when will achieve, Launching vehicle Details, Payloads, How many days mission, what role of this mission, what experiments and studied carried out ,
- Why this Launching Vehicle ?
- Precautions to be taken ?
- Progress Stage till now from beginning
- Proposed Trajectory
- Tell about No. of years to prepare, Total Budget, When, Details of Launching Vehicle, Trajectory, Days to reach , Payloads of Gagnyaan and their role , what are their experiments and images carried out ?

- Name of Scientist, Mission Director, of Gaganyaan, Payload Designer, Where they developed ?
- Gallery of Gaganyaan Images from ISRO
- What Experiments will Carried out – Images
- Curent Status
- Compare with NASA Mission

Reference - Learn from : Documents Uploaded, ISRO Websites , ISRO Websites- About Mangalyaan in details with payload , Launching Vehicle Details – ISDDC Portal and ISRO Website and ISRO Gallery . Internet, Wikipedia , some about Chat GPT – In Chat GPT use IRO Official word in prompt.

Prepare PPT in which slides of each topic with picture , Animated Video . Prepare also word file of it . Learn every point .

Section 13 : Aditya L1

Learn about :

- What is Sun ? Parts or Layers of Sun ? What reactions carried out in Sun ?
- What are the effects of these reactions
- Solar Cycle , Space weather activity like Solar Flares, Radio Blackout, Plasma, CME, Space Weather Parameters
- Effects of Space Weather Activity
- What is Aditya L1 ? Why Halo Orbit ? What is L1 Point . Where it located from earth ? How many hours/minutes/days to reach earth space weather activity to earth
- How Far Sun ? How we reach in Halo Orbit. Complete Trajectory . Number of days ? why we select that day ?
- Role of Aditya L1. What Payloads are used in Aditya L1 in details and Images
- Aditya L1 Launching Vehicle Details, Date of Launch , Why this Date ?
- What we are studying on Sun through Aditya L1 ?
- Tell about No. of years to prepare, Total Budget, When, Details of Launching Vehicle, Trajectory, Days to reach , Payloads of Aditya L1 and their role , what are their experiments and images carried out ?

- Name of Scientist, Mission Director, of Aditya L1, Payload Designer, Where they developed ?

- Gallery of Aditya L1 Images from ISRO
- From where Data of Aditya L1 found ? ISDDC Portal, What Data it give in which format ?
- What Experiments Carried out – Images
- Curent Status
- Compare with NASA Satellites : STEREO, GOES, WIND/DSCOVR and other satellited related to Solar Observations
- Where Centre : India's Solar Observatory and Sun studied ?
- How Aditya L1 better than NASA's satellites of Sun Mission

- **Reference - Learn from : Documents Uploaded, ISRO Websites , ISRO Websites- About Aditya L1 in details with payload , Launching Vehicle Details – ISDDC Portal and ISRO Website and ISRO Gallery . Internet, Wikipedia , some about Chat GPT – In Chat GPT use IRO Official word in prompt.**
- **Prepare PPT in which slides of each topic with picture , Animated Video . Prepare also word file of it . Learn every point .**
- **Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used**

Section 14 : Astrosat & Astronaut

Learn about :

What is Universe, How many Galxies , stars or Celestial Bodies, Multiwavelength space observation, Multiwavelength space observatories details.

cosmic objects—like black holes, neutron stars, and distant galaxies

James Web telescope

Astronomy Mission of India

Astrosat

- Why Astrosat ?

- What we are studying through Astrosats ?
- Tell about No. of years to prepare, Total Budget, When, Details of Launching Vehicle, Trajectory, Days to reach , Payloads of Astrosats and their role , what are their experiments and images carried out ?
- Name of Scientist, Mission Director, of Astrosats, Payload Designer, Where they developed ?
- Gallery of Astrosat Images from ISRO
- From where Data of Astrosat found ? ISDDC Portal, What Data it give in which format ?
- What Experiments Carried out – Images
- Curent Status
- Compare with NASA Satellites /Telescope

Astronaut :

About Astronaut

Astronaut role, eligibility, dress, components, equipments used , threats to astronauts, precautions, astronaut cut out, daily habits, stress management, zero gravity games, how they survive in space ? how they live in space, what they did in space, dress details, past astronauts and their missions carried out by Indian , which organisation sent them by which vehicle, when ?

- **Reference - Learn from : Documents Uploaded, ISRO Websites , ISRO Websites- About Astrosats in details with payload , Launching Vehicle Details – ISDDC Portal and ISRO Website and ISRO Gallery . Internet, Wikipedia , some about Chat GPT – In Chat GPT use IRO Official word in prompt.**
- **Prepare PPT in which slides of each topic with picture , Animated Video . Prepare also word file of it . Learn every point .**
- **Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used**

Section 15 : Launch Vehicle-SLV,PSLV,GSLV

What are Launching Vehicle

Physics used in them

Type of Launching Vehicle

Segments of Launching Vehicle with Details

Fuels used in them and their types and why ? Role of the sections of Vehicle and Fuels

First Launching Vehicle – SLV , design when by whom , Guidance , Failure, Success,

The SLV3 to Mk III journey to LVM III, Cycle to Huge Trailers, 30kg to 4000 kg,

Experimental satellite to Humans, Various parts of LV, Various Fuels,

trajectories, Why SHAR?

About complete Details of SLV, PSLV, GSLV, MK-III and future Launching Vehicles- Complete Details-Role, Segments, Fuels, Separation Stages, From where they launch . Which Missions carried out by which launching vehicle and when ?

- **Reference - Learn from : Documents Uploaded, ISRO Websites , ISRO Websites- About Launching Vehicles in details with payload , Launching Vehicle Details – ISDDC Portal and ISRO Website and ISRO Gallery . Internet, Wikipedia , some about Chat GPT – In Chat GPT use IRO Official word in prompt.**
- **Prepare PPT in which slides of each topic with picture , Animated Video . Prepare also word file of it . Learn every point .**
- **Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used**

Section 16 : Launch Vehicle-LVM III

What are Launching Vehicle

Physics used in them

What is LVM –III ? Compare LVM III with GSLV MK III

Segments of Launching Vehicle LVM III with Details

Fuels used in them and their types and why ? Role of the sections of Vehicle and Fuels

First Launching LVM III Vehicle – LVM III , design when by whom , Guidance , Failure, Success,

The Mk III journey to LVM III,

Experimental satellite to Humans, Various parts of LVM III, Various Fuels, trajectories, Why SHAR?

About complete Details of LVM III and future Launching Vehicles- Complete Details-Role, Segments, Fuels, Separation Stages, From where they launch . Which Missions carried out by which launching vehicle and when ?

- **Reference - Learn from : Documents Uploaded, ISRO Websites , ISRO Websites- About Launching Vehicles in details with payload , Launching Vehicle Details – ISDDC Portal and ISRO Website and ISRO Gallery . Internet, Wikipedia , some about Chat GPT – In Chat GPT use IRO Official word in prompt.**
- **Prepare PPT in which slides of each topic with picture , Animated Video . Prepare also word file of it . Learn every point .**
- **Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used**

Section 17 : RLVTD & Screamjet engine

What are RLVTD & Screamjet Engine

Physics used in them

Segments of these Launching Vehicle with Details

Fuels used in them and their types and why ? Role of the sections of Vehicle and Fuels

RLVTD & Screamjet –design when by whom , Guidance , Failure, Success,

Compare with GSLV.SLV & PSLV Vehicle

About complete Details of RLVTD & Screamjet Engine and future Launching Vehicles- Complete Details-Role, Segments, Fuels, Separation Stages, From where they launch, Trajectories . Which Missions carried out by these launching vehicle and when ?

- **Reference - Learn from : Documents Uploaded, ISRO Websites , ISRO Websites- About Launching Vehicles in details with payload , Launching Vehicle Details – ISDDC Portal and ISRO Website and ISRO Gallery . Internet, Wikipedia , some about Chat GPT – In Chat GPT use IRO Official word in prompt.**
- **Prepare PPT in which slides of each topic with picture , Animated Video . Prepare also word file of it . Learn every point**
- **Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used**

Section 18 : Satellite Components

What is Satellites ? Types ? Various application Types of Satellites

Components and materials used in Satellite- Details, where they import, export , where they developed or integrated , payloads designed and details,

Specs of space hardware, reliability and tests

Various Components and their roles in details .

- **Reference - Learn from : Documents Uploaded, Satellite Communication E Books , ISRO Websites- details with payload Gallery . Internet, Wikipedia , some about Chat GPT – In Chat GPT use IRO Official word in prompt.**
- **Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used**
- **Prepare PPT in which slides of each topic with picture , Animated Video . Prepare also word file of it . Learn every point**

Section 19 : Future Missions

NISAR

- What is NISSAR ?
- What is role ?
- Countries involved.
- Purpose of it.
- Past studied and future plans .
- What we are studying?
- Tell about No. of years to prepare, Total Budget, When, Details of Launching Vehicle, Trajectory, Days to reach , Payloads of NISSAR and their role , what are their experiments and images carried out ?

- Name of Scientist, Mission Director, of NISSAR, Payload Designer, Where they developed ?
- Gallery of NISSAR Images from ISRO
- From where Data of NISSAR found ? ISDDC Portal, What Data it give in which format ?
- What Experiments Carried out – Images

- Curent Status
- Compare with Past studied and Missions by India on it

LUPEX

- What is LUPEX ?
- What is role ?
- Countries involved.
- Purpose of it.
- Past studied and future plans .
- What we are studying?
- Tell about No. of years to prepare, Total Budget, When, Details of Launching Vehicle, Trajectory, Days to reach , Payloads of LUPEX and their role , what are their experiments and images carried out ?

- Name of Scientist, Mission Director, of LUPEX, Payload Designer, Where they developed ?
- Gallery of LUPEX Images from ISRO
- From where Data of LUPEX found ? ISDDC Portal, What Data it give in which format ?
- What Experiments Carried out – Images
- Curent Status
- Compare with Past studied and Missions by India on it

Chandrayaan 4

- What is Chandryaan 4 ?

- What is role ?
- Countries involved.
- Purpose of it.
- Past studied and future plans .
- What we are studying?
- Tell about No. of years to prepare, Total Budget, When, Details of Launching Vehicle, Trajectory, Days to reach , Payloads of Chnadrayaan 4 and their role , what are their experiments and images carried out ?

- Name of Scientist, Mission Director, of Chandrayaan 4, Payload Designer, Where they developed ?
- Gallery of Chandrayaan 4 Images from ISRO
- From where Data of Chandrayaan 4 found ? ISDDC Portal, What Data it give in which format ?
- What Experiments Carried out – Images
- Curent Status
- Compare with Past studied and Missions by India on it

Mangalyaan 2

- What is Mangalyaan 2 ?
- What is role ?
- Countries involved.
- Purpose of it.
- Past studied and future plans .
- What we are studying?

- Tell about No. of years to prepare, Total Budget, When, Details of Launching Vehicle, Trajectory, Days to reach , Payloads of Chnadrayaan 4 and their role , what are their experiments and images carried out ?

- Name of Scientist, Mission Director, of Mangalyaan 2, Payload Designer, Where they developed ?
- Gallery of Mangalyaan 2 Images from ISRO
- From where Data of Mangalyaan 2 found ? ISDDC Portal, What Data it give in which format ?
- What Experiments Carried out – Images
- Curent Status
- Compare with Past studied and Missions by India on it

➢ **Reference - Learn from : Documents Uploaded, Satellite Communication E Books , ISRO Websites and NASA Websites and JAXA - details with payload Gallery . Internet, Wikipedia , some about Chat GPT – In Chat GPT use IRO Official word in prompt.**

➢ **Prepare PPT in which slides of each topic with picture , Animated Video . Prepare also word file of it . Learn every point**

➢ **Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used**

Section 20 : Sounding Rocket

- Concept of Sounding Rocket ,
- Components used in it ?
- Where it demonstrate ? Why Thumba ?
- What studied it done and Why ?

- Purpose of Sounding Rocket ?
- When First time ?
- Atmosphere studies, Niche Apache, TERLS, Magnetic Equator, Every Wednesday – led to excitement and interest,

- Reference - Learn from : Documents Uploaded, ISRO Websites - details with components Gallery . Internet, Wikipedia , some about Chat GPT – In Chat GPT use IRO Official word in prompt.
- Prepare PPT in which slides of each topic with picture , Animated Video . Prepare also word file of it . Learn every point
- **Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used**

Section 21 : Space on Wheel

- What is Space on Wheel ?
- When it starts?
- Its Aim
- What are the feaures and details of it
- Number of Space on Wheels with Region
- How to Schools and Colleges reach out ?
- Try for Extra which You can Learn

- **Reference - Learn from : Documents Uploaded, ISRO Websites - details with components Gallery . Internet, Wikipedia , some about Chat GPT – In Chat GPT use IRO Official word in prompt.**

- **Prepare PPT in which slides of each topic with picture , Animated Video . Prepare also word file of it . Learn every point**
- **Prepare Poster/Chart – Digital as well as Hand Made (By Image Cutting Paste, Manual Draw- Images Used, text Used, Facts Used**